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THE AEROSPACE DEVELOPMENT AND RESEARCH GROUP OF THE UNIVERSIDAD NACIONAL DE COLOMBIA, GIDA-UN, A TOOL FOR AEROSPACE EDUCATION IN COLOMBIA.

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Abstract

The origin and evolution of the Aerospace Research and Development Group of the Universidad Nacional de Colombia, GIDA-UN, for its acronym in Spanish is presented. The group arises in the year 2012 as a student initiative to create an academic space to do activities in the fronts of technology and knowledge appropriation, research in diverse fields in aerospace, and outreach for both the academic community and the general public. This initiative is supported by teachers in the Mechanical and Mechatronics Engineering Department of the Engineering Faculty, and is mainly funded by the Wellness Direction of the Faculty and University. The context in which the group arises is not an easy one. Colombia is a country with low to none governmental or private initiatives in science and technology in general, and in the aerospace field in particular. The aerospace initiatives are in general associated with military and war efforts, and many of the technologies and materials, such as propellants are restricted. At the University, some projects have been developed regarding the topics of rocketry and propulsion, yet they have been punctual initiatives with no continuity, and in general, aerospace initiatives have been discouraged, in relation with the current sociopolitical situation of the country. This initiative aims to fill the gap between the situation it encountered and the interest of students and teachers in the field, while showing the potential this field has in the country. The development of activities in the groups is mostly project based work, with a main project guiding the work, and side projects supporting it or advancing in other fields. The current main project is the Prometeo I mission, which aims to make 4 launches of a sounding rocket to 2km height. The organizational structure of the group is based on that of space agencies, and the work flow is based upon Space Mission Engineering concepts and procedures. The groups work has allowed for the acquisition and spreading of knowledge in the fields of Propulsion, GNC, Mission Design, and field operations, among others. It also has allowed the forming of strategic partnerships with entities that regulate propellant type substances, and provider of test equipment such as a Crawford bomb, and an engine test bench. Outreach activities have also been developed in the form of conferences, workshops and courses for the students, on topics such as propulsion, celestial mechanics and aerospace systems in general.

Keywords: (maximum 6 keywords)

Acronyms/Abbreviations

GIDA-UN: Grupo de Investigación y Desarrollo Aeroespacial- Universidad Nacional de Colombia.

GET: Student Work Group

arisen is the Aerospace Research and Development Group of the Universidad Nacional de Colombia, GIDA-UN, for its acronym in Spanish.

1. Introduction

Access to space, related and spin-off technologies, and derived services is every time more important. The economic impact that the aerospace field has on a worldwide level is growing every year, with several private initiatives appearing every year. As it has been the case for many countries in ways of development, Colombia hasn't had a relevant development on the aerospace field, neither by governmental nor by private initiatives. It is just recently that there have been academic involvements aiming to potentiate the field, as well as startups. One of the academic initiatives that has

This paper aims to present the context on which the group has developed so far and to be the first formal record of the origin and development of GIDA-UN, presenting the past and present technical and educational projects it has executed, and a brief perspective on the future direction of the group activities. It is important to note that some of the events here presented are not formally previously recorded. Also, it is important to notice that this review doesn't aim to be a detailed technical report on the projects that have been developed, but to make a recount of them, their goals, and their development and impact on the group.

2. Colombian Context

Colombia is a country that has endured an internal conflict since the 40's, which has defined the way the country has been developed in the last 60 years. This conflict had his beginnings with junction of several Political and Socioeconomic factors. Through the last 30 years Colombia expended in the military industry according the World Bank approximately 3,2% of the GDP. [1]

Colombian government has mainly invested this large amount of money with the purpose of end the internal conflict and maintain his sovereignty. It had fought with many armed actors to do so. These are belligerents which fight between them and against Colombia's militar force; like AUC, FARC-EP, ELN, ELP, and common delinquency. Among them FARC-EP is the biggest armed group in Colombia which is currently estimated to have 13800 members. [2] As mentioned before Colombia is investing a great percent of his GDP in maintain the peace, but for doing so, it has forgotten others possible sectors that can be develop by investing a fraction of this percentage. But since 2012 Colombian government began the peace dialogues in La Habana with FARC-EP. These dialogues ended in august of the current year, with a final peace agreement between the two parts involved. Looking this other armed actors had beginning the dialogues with Colombian government. All the money that Colombian government was investing fighting against them will be available. Thus this agreement sets up a new era for Colombia, because the resources allocated for war efforts could be used in other sectors, like education, or Science and technology. Nowadays the last sector has less than the 0,25% of the GDP according to the World Bank, but could grew up by redirecting the resources. Therefore Colombia has a variety of fields in science and technology to choose in which ones is going to invest. Possible the government could spend this recently freed money in the aerospace field instead of enhancing his military Air Force.

Consequently with the low expenditure in research and development of science and technology, Colombia is a country which economy is based in commodities and natural resources extraction. Hence the primary economic activities of Colombia are Energy and Mining, and Agriculture. Both of them have been suffering through the last years. The first one because the decreasing of fossil fuel international prices and the environmental effects in his extraction, this has behave erratically meanwhile the Agriculture had been affected by the high-tech competition, which produce more and cheaper than Colombia's farmers. Therefore Colombia needs to find new economic activities, this can be

achieve investing the old resources of war in a good way. Like in Science and technology field. [1]

3. Theory and calculation

The first approach to aviation in Colombia was in the year 1911, when several businessman hired Paul Miltgen with the aim to made the first air show in Colombia, this was intended to happen in Bogota, but the high altitude was not planned and the results were not very prosperous. [3] This shows that to make any aerospace event or development, is mandatory to take in count every single variable.

In the year 1921 two air companies were founded, the first one, but not to famous, was the CCNA (Compañía Colombiana de Navegación Aérea), the second one was SCADTA (Sociedad Colombo-Alemana de Transportes Aéreos). These two companies only flew to a small part of Colombia. Between the two world wars were founded 9 Air companies that help to develop some cities like Villavicencio, which became the door of an entire region of Colombia[3] showing how the aerospace in Colombia can lead to develop the country. The Commercial Aviation achieved a big growth after the development of military aviation for the world war II. This growth is shown in the at least 18 Air companies that were founded in less than 10 years.[3] Colombia has the oldest Air Company in America and nowadays in Colombia are 158 certificated Air commercial companies according the "Aeronautica Civil" [4]

In the year 2006 the government created the Colombian Space Committee (CCE). This Committee orients the implementation of the Colombian national policies for the development and application of space technologies, also the development of plans, programs and projects in this field. His aim is to maximize the contribution of the aerospace field to social, economic and cultural development of Colombia. [5]

In Colombia mainly two aerospace private initiatives exist, Sequoia Space [6] and the Program "Medellín Espacial" [7]. The first one is aiming to develop small and nanosatellites, the second one try to show that through Space is possible to create and promote new ventures in this new economic sector. In the academic sector several Universities have been studied about this field, but none of them had realized a continuous work.

For example the Universidad Nacional de Colombia, more specifically the Mechanical and Mechatronics Engineering Department in his 50 years had done only three works about the aerospace field, which shows that the interest of this subject is not much. As a result of this low interest, there has not been a continuous

process. [8] Other punctual initiatives in the Universidad Nacional de Colombia had been the following: A book of Propulsion Technology. Introducción general a la tecnología de propulsión / Diego Alexander Garzón Alvarado, Carlos Alberto Duque Daza, Máximo Alejandro Roa Garzón. A prototype of a liquid fuel micromotor Micromotor cohete : desarrollo de un prototipo experimental para combustible líquido / Juan Miguel Contreras Bustamante ; dir. Gabriel Meluk Orozco archivo de computador. And the study of the trajectory of three stage Rocket in Colombia. Estudio de la trayectoria de un cohete de tres etapas lanzado desde el territorio colombiano / Jhonathan Orlando Murcia Piñeros ; director José Gregorio Portilla Barbosa.

It is also important to mention that other aspects of the aerospace field that have been explored correspond to Astrobiology and Planetary Sciences, with groups constituted at the Science Faculty, and which with whom projects have been conducted.

4. History of GIDA-UN

The beginning of the group can be traced to early 2012 when a group of students reunited with the intention of start a work related to the aerospace field. This beginning was not started with a formal project, but with self-instruction sessions and collaboration on the calibration on a small wind tunnel which at the moment was located at the mechatronics laboratory of the university. The proper organization of the group began in the first semester of 2013. It is important to note the figure under which it was created. The Wellness Direction of the university offers a figure under which students can organize around a project, which is presented under the Logical Framework Analysis, this figure is called Student Working Groups, GET for its acronym in Spanish (Grupo Estudiantil de Trabajo). The group has to reunite certain requirements for its formation, among others, it has to be constituted of at least 3 students of the same faculty, and a director teacher. The areas which can be considered for GET's are cultural and aesthetic contributions, free time and ludic activities, publications, human and community development, and academic contribution, which is where GIDA-UN is classified. Once the proper documentation is presented, the proposed project is evaluated by committees consisting of faculty and university directives and a budget is assigned, according to the stated requirements. The process of validation is made on a semestral base, with new projects being presented, or extensions requested. At the end of the term, a report on the activities is presented with a written report and an oral presentation.

From then on, the group has presented projects, and has had continuous work. It is important to note that according to data supplied by the Wellness Direction, in a period from 2008 to 2014, 63% of the groups created in the Engineering Faculty only lasted one semester, and 86% lasted 3 semesters or less. The specific causes of the short duration of the initiatives remain to be determined, but one of them is associated with the conforming students graduating, leaving no one to follow up. Regarding this issue, the group has started an initiative called *semillero* which will be further presented.

The group was created with two goals in mind, the first one is to learn and make a technology appropriation process, relating to the different aspects of aerospace technology, since the creation of qualified personnel and the mastering of techniques is a key aspect for the success of space development. The second one is related to the creation of outreach programs, to communicate different aspects of the aerospace field to the academic community, within the university, and to the society in general, since the support of the local communities is important for the programs to be developed, and the general scientific illiteracy is an obstacle in that sense. GIDA-UN has grown in organization, according to the needs it has encountered as the projects grew in complexity and reach. It is also important to note that the Wellness Direction does not require for the groups to be internally organized in any specific way, apart from having a director teacher and a coordinator, which has to be one of the students.

The group is directed by the teacher and the coordinator, the rest of the students are organized in teams, defined by area of work, each team has a leader which responds directly to the coordinator, this allows for a clear decision making path, and a quick information flow. The teams are:

Propulsion: In charge of the development of the fuel and rocket engines, as well as the integration with the structure.

Avionics: Devoted to the development and testing of the GNC, and electronic systems for the different applications, such as vehicles, ground station, and test equipment.

Structure & Aerodynamics: In charge of designing, testing and manufacturing the structure of the devices, taking into account aerodynamic considerations. It is also in charge to coordinate with the other teams as an integration point.

Recovery: Devoted to the design and testing of the recovery systems for the different vehicles.

Outreach: In charge of handling the interaction with the media, the outreach programs, and to manage the communication channels and social networks.

Management: In charge of the administrative tasks, the implementation of project management programs, inter-institutional interactions, requesting permissions, task assignments

The teams are not separated by majors, because an interdisciplinary work is promoted, and it allows for several points of view are presented. While the majority of the members of the group are, and have been, from engineering and sciences, there is participation of students from other areas of knowledge, such as industrial design, administration, and cinema and television.

Mainly, the budget of the group is provided by the Wellness Direction and the Engineering Faculty. The budget approved for the periods from 2013 to 2015 are presented in Fig. 1. The values are presented in millions of Colombian Pesos, taking into account that as of 2016 1USD=3000COP. It is also important to note that there has been a component of autofinancing, coming from the students, that is not taken into account, for it is hard to trace, but it is around 15% of the group's budget. Also, transportation and participation in events are not accounted for.

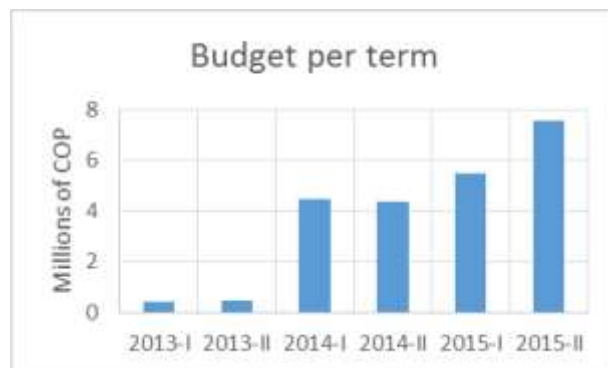


Fig. 1: Budget per term.

The projects that have been developed by the group so far are:

Water rockets: Given the restriction of oxidizers, and launch places, the place to start for the group was with water rockets. The relatively simple devices were used as a platform to develop an understanding on basic rocketry concepts, such as aerodynamics and propulsion. Later, some rockets were improved, as well as its launching pads, to uphold higher pressures, and a cargo bay was adapted in order to equip the rockets with a first form of telemetry and remotely operated recovery

systems. Currently the developments in water rockets are still being used as an educational and demonstrative tool, as an example in the crew training for the launch team activities, and as a testing method for some of the instruments for the more evolved rockets.



Fig. 2: Outreach activity involving the groups water rocket platform.

OpenROV: In order to appropriate the knowledge about remotely operated vehicles, and their working protocols, the OpenRov, an open source remotely operated submarine was acquired. The vehicle was assembled and set up for running. however, due to the lack of continuity on a proper space for testing the device, it was lent to another group which continued the study and development.



Fig. 3: GIDA-UN's Assembled OpenROV.

Simte I: while not intended for flight, the Simte I was the first iteration of the group in designing solid fuel rockets. The project served as a base to identify the different subsystems that should be taken into account for the correct development and execution of this type of project. The project aimed to design a rocket for a 1km apogee. The design considered also telemetry and recovery systems, as well as basic aerodynamic

considerations for subsonic flight. It is important to mention that the current team organization of the group was largely influenced by the development of this project.



Fig. 4: Simte I Rocket in CAD.

The development of the aforementioned projects, and also of the current mission, which will be described next, has allowed to associate with key players in the aerospace field in Colombia, among them, the Colombian Space Committee, the Colombian Air Force, which through the Space Affairs Office has lend help to the planning of the Prometeo I mission, Indumil, a governmental company dedicated to the manufacturing and control of guns, ammunition, and precursor substances, and other universities, which are interested in the development on joint projects.

5. Current Projects

As it was described before, GIDA-UN's work is mainly guided by a main project, and a series of side project which might or not be related to the first one. This is made with two objectives, the first one is to advance technically in several fronts, allowing for a growth in the general knowledge base, as well as a way to acquire technologies or equipment which can facilitate the main project. The second goal is to respond to the curiosity, expectations and personal areas of interest of the members of the group, and of each team.

The current main project is the Prometeo I mission, which will be the first experience with solid fuel rockets in the group. The mission arises from the necessity of working with more advanced technology that of what had been already done. The mission goals are presented in two ways, first, a set of technical goals which are to launch and recover 2 Simte II rockets, 2 times each, to a height of at least 2km above the launching point, equipped with telemetry instruments and recovery systems, and with the ability to hold payload. This mission architecture was selected in order to have parameters that allow for the quantification of manufacture quality and repeatability of the obtained results. The second set of goals is organizational, and aims to determine and acquire the required abilities and

equipment in order to be able to properly design, evaluate, test, manufacture and launch a rocket that accomplishes the technical specifications mentioned before.[9]

The execution of this mission involved a series of challenges, given that at the beginning of the project there was no practical expertise and little theoretical knowledge in topics like manufacturing and handling rocket fuel, launch protocols, GNC equipment or software, test equipment, mission design and project management.

Some advancements of this project were presented at the VIII Congreso Argentino de Tecnología Espacial, CATE, held at Buenos Aires on may 2015. Between the presentations were the design of the mission, which was made taking into account Space Mission Engineering concepts, and the development of the procedure for manufacturing the solid fuel required for the mission.

The development of the mission has led to developing and refining technology in engines, with a refined, yet manual technique for solid fuel grains manufacturing, the development and testing of telemetry, control and communications module, it has led to an interaction with local suppliers and manufacturers for certain parts to be prepared, and to development of protocols for test and launch procedures, such as the engine testing presented next in fig.5.



Fig. 5: Engine testing at Indumil Explosive Plant FEXAR testing facilities.

Among this procedures, security is top priority, a uniform for the members of the launch team is required, consisting of an orange overall, with 4 identifications, GIDA-UN's logo, the Universidad Nacional de Colombia logo, Mission patch, and name with blood

type. The uniform is complemented with a helmet, with a color code identifying the team and the chain of command. Radio units are used for communication on field operations, with prestablished commands and calls. This protocols have been put to test in activities such as engine and communication testing, and trainings, some of which take place in arid places, similar to the environment of the launch ground.



Fig. 6: Training at Tatacoa Desert, Huila, Colombia.

As part of the activities, a mission patch was designed, in order to give identity to the initiative, and to the members of the group. The patch is presented next, in fig. 7.



Fig. 7: Prometeo I Mission Patch.

The launches of the mission are expected to take place on October 16 2016 at Villa de Leyva, a municipality at about 160km from Bogotá D.C. The place was selected due to its arid environment, low to none air traffic, and a low population.

Apart from the main project, side projects will be briefly mentioned, and they include:

Test equipment: Together with Indumil, a test bench for rocket engines is being tuned. The bench is built with a capacity for thrust up to 25000N. The data capturing software is being tested and the respective sensors calibrated. A centrifugal acceleration calibration bench is also being calibrated, as well as a vibration device, to set to test the different electronic devices and payloads that are going to be on board the launch vehicles.

Rover: A rover type vehicle is currently being characterized, set up and prepared for instrumentation that will aid field exploration in zones of astrobiology or geological interest. The first stage of the project contemplates to study the dynamics of the vehicle, to measure its capabilities and to mount instruments appropriate for the field work it is intended to serve. The second stage aims to make simulated excursions to properly calibrate the instruments, and to refine the developed protocols of interaction with the vehicle. The third stage aims to implement the system in a real environment, to aid with the characterization of a certain area of interest.

Martian Atmosphere Simulator: In conjunct work with the Astrobiology and Planetary Sciences Group of the University, a first prototype of a simulation chamber for Mars conditions is being developed. The system aims to simulate pressure, gas mixture, and UV radiation on the surface of Mars, in order to test the viability of terrestrial extremophiles.

Atmospheric balloon: in order to test telemetry and communication systems for higher altitude rockets, a balloon is being developed. The goal is to ensure the ability to communicate with the next generation of launch vehicles from ground control. For this, antennas have been selected and a series of tests proving the software and the communication protocols was conducted on a distance of approximately 2,5km.

6. Education Projects

As it was mentioned before, one of the main goals of the group is related to the educational projects, as it is the base for motivation for people to enter a related field, to increase the scientific literacy of the community and to create a base of knowledge to increase the advancement on the field. The first aspect in which the group contributes to the education is through the execution of the projects themselves. The majority, not to say all, of the members are students, and a great majority are undergrads. The involvement of this population allows for them to apply knowledge gained in the subjects, which sometimes is left as a theoretical aspect, as well as promoting interaction, team work, and the

implementation of self instructing and problem solving habits, in an environment where there is certain error tolerance is accepted.

Direct outreach campaigns have also been developed, with 4 formal series of conferences organized, as well as forums and in general activities where the public has access to certain topics on the aerospace field. Among the conferencists have been students of the group, experts from Airbus Defence and Space, students from other universities, local teachers with interest in the field, and representatives from the industry. Among this events, a one day workshop type event was organized with the Space Generation Advisory Council, within a larger event, named Space Week GIDA-UN 2015. In this event, the topics were Space Mission Simulations, Applications of Astrobiology in Colombia, and Education on science and the role of the aerospace field. Mass media attention on the activities has also appeared, with the group having appeared in two national TV Channels, National Radio, National newspapers, and one local channel. The group also uses Facebook fanpages, Twitter, and Youtube to promote its activities, and general news on the aerospace field.

Another initiative which its pilot is in course is the implementation of educational missions with basic and high schools. The program is currently in course with the Colegio Ricaurte (Ricaurte School), consisting on giving a one day workshop to the students, on basic concepts of aerospace science and technology. Afterwards they develop a payload project, which has specific requirements to fit on the Simte II payload bay. After being selected and tested for possible interference with the vehicles flight, the experiment is launched and recovered, in order for the students to evaluate the results. It is expected for this exposure to a real launch incentives the interest for the field, and allows them to experience firsthand the technical process of a payload development.

As it was mentioned before, one problem for the disappearance of student groups was the fact of some of the members graduating, leaving no one to follow up. To attack this problem a course denominated *semillero*, which would translate to seedbed, was created, aimed for students between 1st and 4th semester, it provides the participants with insights related to the aerospace field, that generates interest in them, and at the same time allows them to enter the group with a conceptual knowledge on the topics.

Currently, and taking into account the lack of bibliography in Spanish, a book is being developed, with aims to be published in 2017. The book will be

titled “An introduction to Aerospace Sciences and Engineering” in Spanish, and will cover a variety of topics, including history, propulsion, celestial mechanics, environmental control and life support systems, and GNC, among others.

7. Future Projects

The range of projects, interests and ambition in the group is rapidly growing. Some of the future projects that are currently in the planning stage are going to be mentioned. It is important to note that while the specificities of the projects may change, the work lines are to be explored.

Launching vehicles: It is expected that with a better knowledge, obtained from the experience with the Prometeo I mission, a new generation of launching vehicles is to be designed. The first proposed project is the Prometeo II mission, with a tentative goal of 25km and a 5kg payload. As with the first Prometeo mission, this one would also have organizational objectives, and being more ambitious, it aims to generate an appropriate work flow between different actors involved, with the intention to interact with other universities, local industry, government, and media, with outreach programs.

Exploration Vehicles: With the knowledge derived from the experiences with the Open ROV and the Rover, a next generation of exploration vehicles is expected to be developed. It is expected for them to take into account parameters like usability of certain materials and techniques depending on the environmental conditions of the planetary body it is expected to explore.

Human Performance: The development of devices that aid human exploration is desired. Equipment for field location, and human-machine interaction with the goal of exploration is to be developed. Also, training for different launch activities, and the reduction of human error is desired.

Test Equipment: A next generation of testing equipment is to be developed, improving the capabilities of the ones already implemented. It is also desirable to develop or acquire test equipment in fields that are not already implemented.

Satellites: A first approach to small satellite implementations is to be developed. A first phase would search for the knowledge appropriation, seeking to also identify missions that can be covered with the current capabilities. A second phase would aim to implement the manufacturing of a small satellite with national

manufacture, in order to test systems and subsystems, being a technology demonstration device. Finally, a satellite with a scientific, commercial or governmental mission would be developed.

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PER ASPERA AD ASTRA...!

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